

Geometric ML Pipeline: From Market Data to Unsupervised Crisis Detection

Financial Time Series

*Market returns
(5 ETFs, 17 years)*



Hilbert Space Embedding

*PCA $\rightarrow$ quantum
state vectors $|\psi(t)\rangle$*



Geometric Observables

*Berry curvature
QFI $\cdot$ Fidelity*



Regime Detection

*Crisis alerts
(no labels needed)*